

Dataset & Novelty Description for Research Paper

1. Overview

This research presents an India-wide Crop Recommendation System using real agricultural data from 17 years (2000–2017)

across 550+ districts. Unlike traditional models that rely only on NPK and climatic factors, our system integrates

actual crop yield data from ICRISAT with district-wise soil micronutrient data.

2. Datasets Used

1. ICRISAT District-Level Agriculture Dataset (2000–2017)

■ Source: <https://data.icrisat.org>

■ Government-supported dataset

■ Data includes Area, Production, Yield for major crops across 550+ districts

2. Soil Nutrient Dataset (Kaggle – Non-Govt.)

■ Source: <https://www.kaggle.com>

■ Micronutrients used: Zinc (Zn), Iron (Fe), Copper (Cu), Manganese (Mn), Boron (B), Sulfur (S)

3. Kaggle Crop Recommendation Dataset (Baseline)

■ Source: <https://www.kaggle.com/atharvaingle/crop-recommendation-dataset>

■ Features: N, P, K, rainfall, pH, temperature, humidity

■ Purpose: Used only for baseline comparison

3. Novelty – What Makes This Research Unique?

■ Uses real-world yield data, not simulated/lab-based values

■ Covers 17 years & 550+ districts → region-aware crop prediction

■ Merges soil micronutrients + historical yield = practical recommendation

■ Baseline vs Real-World Comparison: Kaggle dataset vs ICRISAT dataset

■ Suitable for Indian conditions → can be used by farmers/government apps

4. Why This Approach Makes a Difference?

- Traditional Kaggle models ignore district & crop history → we include both
- We use real farmer-recorded production and area data, not assumptions
- Crop suitability changes across states — our model learns this using actual data
- Micronutrients (Zn, Fe, Cu, Mn, B, S) improve precision over NPK-only models

5. Final Dataset Summary

- Total Records: 77,762
- Time Span: 2000–2017
- Geographic Coverage: 550+ districts, 29 states
- Columns: Year, State, District, Crop, Area, Production, Yield, Zn, Fe, Cu, Mn, B, S, district_key

6. ML Goal

Input (X): Soil nutrients + State + District + Year + Yield

Output (y): Most suitable Crop (Classification)

7. Links

- ICRISAT Data: <https://data.icrisat.org>
- Soil Data: <https://www.kaggle.com>
- Kaggle Crop Dataset: <https://www.kaggle.com/atharvaingle/crop-recommendation-dataset>